

8. (a) The following table gives information about some atoms, **A-E**.

A-E are not the chemical symbols of the elements.

Atom	Number of protons	Number of neutrons
A	12	12
B	10	10
C	6	8
D	11	12
E	6	6

(i) Give the letter, **A-E**, of the atom with a mass number of 12. [1]

.....

(ii) Give the letter, **A-E**, of the atom found in Group 1 of the Periodic Table. [1]

.....

(iii) Choose the letters, **A-E**, of the atoms that are isotopes. Give the reason for your choice. [2]

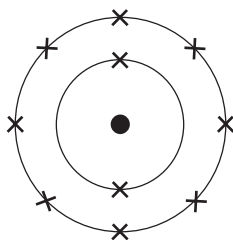
Letters and

Reason

.....



(b) The diagram below shows the electronic structure of an element in the Periodic Table.



(i) Using $\times$ to represent an electron, draw a diagram to show the electronic structure of the element which lies directly **below** this one in the Periodic Table. [1]

(ii) Explain why both of these elements are chemically unreactive. [1]

.....

.....

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



Group

2

→

24



${}^1_1\text{H}$ Hydrogen

1 H Hydrogen 1														
7	Li	9												
	Lithium	Be												
	3	Beryllium												
		4												
23	Na	24												
	Sodium	Mg												
	11	Magnesium												
		12												
39	K	40												
	Potassium	Ca												
	19	Calcium												
		20												
86	Rb	88												
	Rubidium	Sr												
	37	Strontium												
		38												
133	Cs	137												
	Caesium	Ba												
	55	Barium												
		56												
45	Sc	48												
	Scandium	Ti												
	21	Titanium												
		22												
51	V	52												
	Vanadium	Cr												
	23	Chromium												
		24												
55	Mn	56												
	Manganese	Fe												
	25	Iron												
		26												
59	Co	59												
	Cobalt	Ni												
	27	Nickel												
		28												
63.5	Cu	65												
	Copper	Zn												
	29	Zinc												
		30												
103	Rh	106												
	Rhodium	Pd												
	45	Palladium												
		46												
101	Ru	112												
	Ruthenium	Cd												
	44	Cadmium												
		48												
99	Tc	108												
	Technetium	Ag												
	43	Silver												
		47												
184	W	197												
	Tungsten	Au												
	74	Gold												
		79												
181	Ta	201												
	Tantalum	Hg												
	73	Mercury												
		80												
179	Hf	195												
	Hafnium	Pt												
	72	Platinum												
		78												
186	Re	201												
	Rhenium	Hg												
	75	Mercury												
		80												
190	Os	201												
	Osmium	Hg												
	76	Mercury												
192	Ir	201												
	Iridium	Hg												
	77	Mercury												
		80												
195	Pt	201												
	Platinum	Hg												
	78	Mercury												
197	Au	201												
	Gold	Hg												
	79	Mercury												
		80												
204	Tl	207												
	Thallium	Pb												
	81	Lead												
		82												
207	Pb	207												
	Lead	Pb												
	82	Lead												
209	Bi	209												
	Bismuth	Po												
	83	Polonium												
		84												
210	Po	210												
	Polonium	Po												
	84	Polonium												
210	At	210												
	Astatine	At												
	85	Astatine												
222	Rn	222												
	Radon	Rn												
	86	Radon												

11	B	12	C	14	N	16	O	19	F	20	Ne
	Boron		Carbon		Nitrogen		Oxygen		Fluorine		Neon
	5		6		7		8		9		10
27	Al	28	Si	31	P	32	S	35.5	Cl	40	Ar
	Aluminium		Silicon		Phosphorus		Sulfur		Chlorine		Argon
	13		14		15		16		17		18
70	Ga	73	Ge	75	As	79	Se	80	Br	84	Kr
	Gallium		Germanium		Arsenic		Selenium		Bromine		Krypton
	31		32		33		34		35		36
115	In	119	Sn	122	Sb	128	Te	127	I	131	Xe
	Indium		Tin		Antimony		Tellurium		Iodine		Xenon
	49		50		51		52		53		54
204	Tl	207	Pb	209	Bi	210	Po	210	At	222	Rn
	Thallium		Lead		Bismuth		Polonium		Astatine		Radon
	81		82		83		84		85		86

Key

A_r	Symbol	Name	Z
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relative atomic mass

name	atomic number
7	

Surname	Centre Number	Candidate Number
Other Names		0



GCSE – **NEW**

3430U20-1



SCIENCE (Double Award)

Unit 2: CHEMISTRY 1
FOUNDATION TIER

WEDNESDAY, 13 JUNE 2018 – MORNING

1 hour 15 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	7	
2.	9	
3.	7	
4.	6	
5.	10	
6.	6	
7.	8	
8.	7	
Total	60	

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question **6** is a quality of extended response (QER) question where your writing skills will be assessed.
The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.



JUN183430U20101

8. The table shows some physical properties of Group 6 elements.

Element	Melting point (°C)	Boiling point (°C)	Density (g/cm ³)	Electrical conductor
oxygen	−219	−183	0.0014	no
sulfur	115	445	2.0	no
selenium	221	685	4.8	semi-conductor
tellurium	450	988	6.2	semi-conductor

(a) (i) Describe the trend in the melting points of the Group 6 elements. [1]

.....

(ii) Give the physical state of selenium at 400 °C. Give a reason for your choice. [2]

.....

.....

(iii) Explain why it is difficult to classify selenium as either a metal or a non-metal. [1]

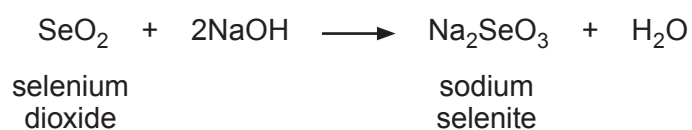
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(b) Selenium dioxide reacts with sodium hydroxide to produce sodium selenite and water.



(i) Calculate the relative formula mass (M_r) of sodium selenite, Na_2SeO_3 . [1]

$$A_r(\text{Na}) = 23 \qquad A_r(\text{Se}) = 79 \qquad A_r(\text{O}) = 16$$

$M_r = \dots\dots\dots$

(ii) Calculate the percentage by mass of selenium in sodium selenite. [2]

Percentage = $\dots\dots\dots$ %

7

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



THE PERIODIC TABLE

Group 1 2 3 4 5 6 7 0



24

¹ H Hydrogen 1

⁷ Li Lithium 3	⁹ Be Beryllium 4
²³ Na Sodium 11	²⁴ Mg Magnesium 12
³⁹ K Potassium 19	⁴⁰ Ca Calcium 20
⁸⁶ Rb Rubidium 37	⁸⁸ Sr Strontium 38
¹³³ Cs Caesium 55	¹³⁷ Ba Barium 56
²²³ Fr Francium 87	²²⁶ Ra Radium 88
	²²⁷ Ac Actinium 89

¹¹ B Boron 5	¹² C Carbon 6	¹⁴ N Nitrogen 7	¹⁶ O Oxygen 8	¹⁹ F Fluorine 9	²⁰ Ne Neon 10
²⁷ Al Aluminium 13	²⁸ Si Silicon 14	³¹ P Phosphorus 15	³² S Sulfur 16	^{35.5} Cl Chlorine 17	⁴⁰ Ar Argon 18
⁷⁰ Ga Gallium 31	⁷³ Ge Germanium 32	⁷⁵ As Arsenic 33	⁷⁹ Se Selenium 34	⁸⁰ Br Bromine 35	⁸⁴ Kr Krypton 36
¹¹⁵ In Indium 49	¹¹⁹ Sn Tin 50	¹²² Sb Antimony 51	¹²⁸ Te Tellurium 52	¹²⁷ I Iodine 53	¹³¹ Xe Xenon 54
²⁰⁴ Tl Thallium 81	²⁰⁷ Pb Lead 82	²⁰⁹ Bi Bismuth 83	²¹⁰ Po Polonium 84	²¹⁰ At Astatine 85	²²² Rn Radon 86

⁵⁵ Mn Manganese 25	⁵⁶ Fe Iron 26	⁵⁹ Co Cobalt 27	⁵⁹ Ni Nickel 28	^{63.5} Cu Copper 29	⁶⁵ Zn Zinc 30
⁵² Cr Chromium 24	⁵¹ V Vanadium 23	⁴⁸ Ti Titanium 22	⁹¹ Zr Zirconium 40	⁹³ Nb Niobium 41	⁹⁶ Mo Molybdenum 42
⁹⁹ Tc Technetium 43	¹⁰¹ Ru Ruthenium 44	¹⁰³ Rh Rhodium 45	¹⁰⁶ Pd Palladium 46	¹⁰⁸ Ag Silver 47	¹¹² Cd Cadmium 48
¹⁸⁶ Re Rhenium 75	¹⁹⁰ Os Osmium 76	¹⁹² Ir Iridium 77	¹⁹⁵ Pt Platinum 78	¹⁹⁷ Au Gold 79	²⁰¹ Hg Mercury 80
¹⁸¹ Ta Tantalum 73	¹⁷⁹ Hf Hafnium 72	¹⁸⁴ W Tungsten 74			

Key

^{A_r} Symbol Name Z	relative atomic mass	atomic number
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Surname	Centre Number	Candidate Number
Other Names		0



GCSE

3430U20-1



WEDNESDAY, 12 JUNE 2019 – MORNING

SCIENCE (Double Award)

Unit 2: CHEMISTRY 1
FOUNDATION TIER

1 hour 15 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
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2.	8	
3.	6	
4.	7	
5.	6	
6.	6	
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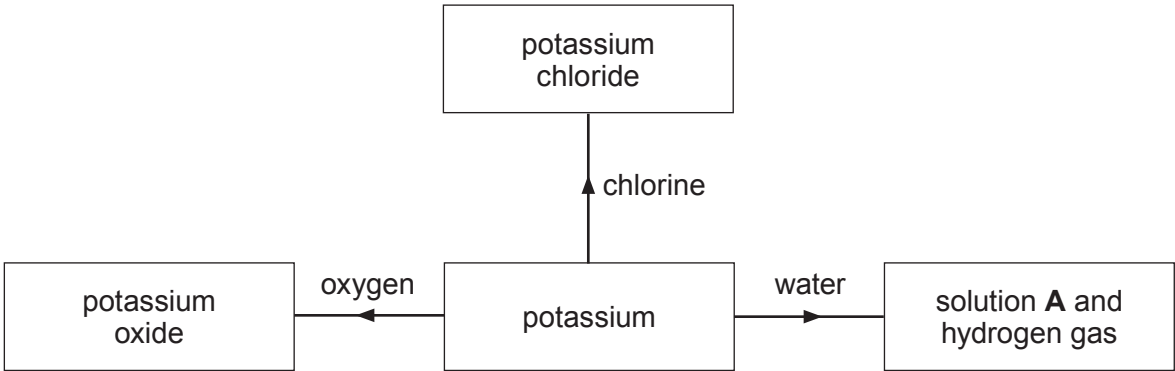
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8. (a) The table gives information about some elements.

Element	Electronic structure	Group	Period
oxygen	2,6	6	2
chlorine		7	3
.....	2,8,5	5	3
potassium	2,8,8,1	1	

Complete the table. [3]

(b) The flow chart shows some of the reactions of potassium.



(i) State **one** observation you would make when potassium reacts with water. [1]

.....

.....

(ii) Apart from wearing gloves and safety goggles, give **one** safety precaution that should be taken when adding potassium to water. [1]

.....

.....



(iii) Give the formula of solution **A**. [1]

.....

(iv) Suggest a value for the pH of solution **A**. [1]

.....

(v) Name a Group 1 metal that is **more** reactive than potassium. [1]

.....



8. (a) (i) State why sodium is stored in oil in the laboratory. [1]

.....

(ii) Describe the change in appearance when a piece of freshly cut sodium is left for a few minutes. [1]

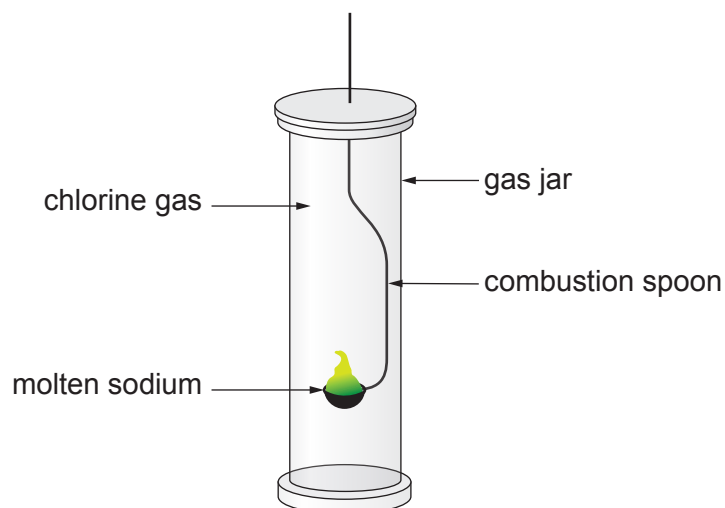
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(iii) Give the formula of the compound formed when sodium reacts with oxygen. [1]

.....

(b) The diagram shows the reaction of sodium with chlorine.



(i) State why it is necessary to carry out this reaction in a fume cupboard. [1]

.....

(ii) Complete and balance the equation for the reaction of sodium and chlorine. [2]



Examiner
only

(c) The table shows some properties of Group 7 elements.

Element	Melting point (°C)	Boiling point (°C)	Reaction with hot iron
fluorine	−220	−188	explosive
chlorine	−101	−34	very fast
bromine	−7	59	quite fast
iodine	114		slow

(i) Put a tick (✓) in the box next to the most likely boiling point for iodine. [1]

−25 °C ☐ 25 °C ☐ 100 °C ☐ 150 °C ☐

(ii) Astatine lies below iodine in Group 7. State how you would expect astatine to react with hot iron.

Give a reason for your answer. [2]

.....

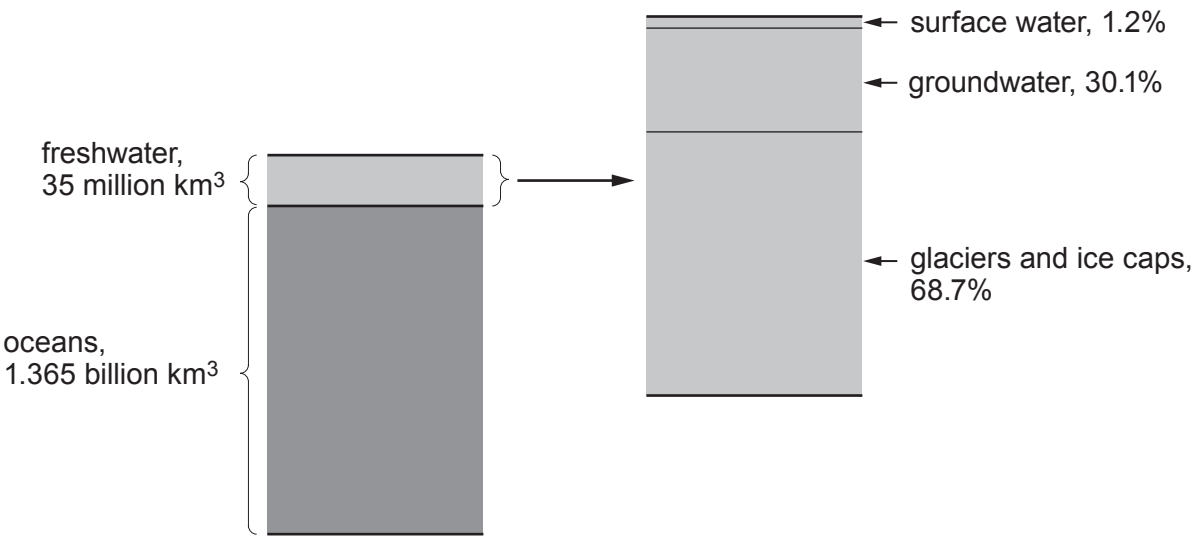
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9



8. The diagram shows the distribution of water on Earth.



(a) Use the information in the diagram to calculate the volume of freshwater that is available as surface water. [2]

Volume = million km³

(b) As the Earth’s population continues to increase, there is going to be a greater need to get freshwater from seawater. This is known as desalination.

Give a method that could be used to obtain freshwater from seawater. [1]

.....



Examiner
only

(c) Calcium ions, Ca^{2+} , cause hardness in groundwater.

Describe an experiment to **compare** the hardness of two groundwater samples. [3]

.....

.....

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.....

(d) Washing soda is used to soften hard water. This contains sodium carbonate.

Give the formula of sodium carbonate. [1]

.....

7

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
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calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



Group

2

no

5

13

1 H Hydrogen 1																																			
7	Li	9	Be							11	B	12	C	14	N	16	O	19	F	20	Ne														
	Lithium		Beryllium																																
23	Na	24	Mg																																
	Sodium		Magnesium																																
39	K	40	Ca	45	Sc	48	Ti	51	V	52	Cr	55	Mn	56	Fe	59	Co	59	Ni	63.5	Cu	65	Zn	70	Ga	73	Ge	75	As	79	Se	80	Br	84	Kr
	Potassium		Calcium		Scandium		Titanium		Vanadium		Chromium		Manganese		Iron		Cobalt		Nickel		Copper		Zinc		Gallium		Germanium		Arsenic		Selenium		Bromine		Krypton
86	Rb	88	Sr	89	Y	91	Zr	93	Nb	96	Mo	99	Tc	101	Ru	106	Pd	108	Ag	112	Cd	115	In	119	Sn	122	Sb	127	Te	128	I	131	Xe		
	Rubidium		Strontium		Yttrium		Zirconium		Niobium		Molybdenum		Technetium		Ruthenium		Palladium		Silver		Cadmium		Indium		Tin		Antimony		Tellurium		Iodine		Xenon		
133	Cs	137	Ba	139	La	179	Hf	181	Ta	184	W	186	Re	190	Os	192	Ir	195	Pt	197	Au	201	Hg	204	Pb	207	Bi	209	Po	210	At	222	Rn		
	Caesium		Barium		Lanthanum		Hafnium		Tantalum		Tungsten		Rhenium		Osmium		Iridium		Platinum		Gold		Mercury		Lead		Bismuth		Polonium		Astatine		Radon		
223	Fr	226	Ra	227	Ac	Key										227	Fr	228	Ra	229	Ac														
	Francium		Radium		Actinium																														

Key

relative atomic mass

A_r	Symbol	Name	Z
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atomic number